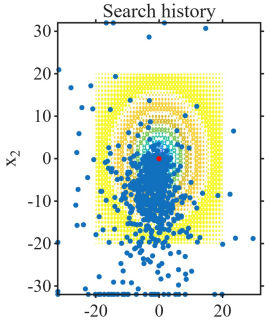
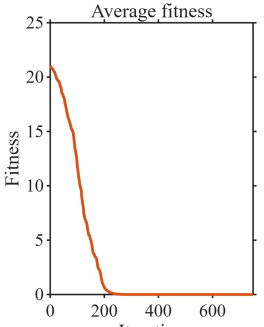
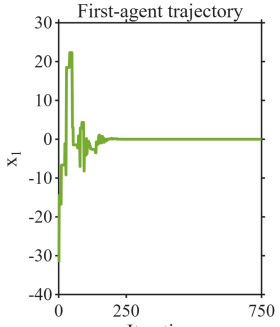
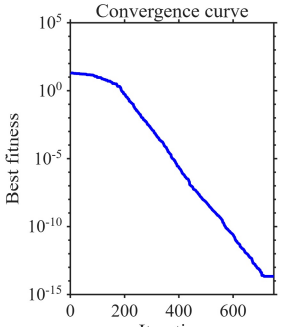

F10
Fitness
20
15
10
5
20
0
-20
-20
0
20
x2
x1


Search history
x2
30
20
10
0
-10
-20
-30
-20
0
20
x1


Average fitness
Fitness
25
20
15
10
5
0
0
200
400
600
Iteration


First-agent trajectory
x1
30
20
10
0
-10
-20
-30
-40
0
250
750
Iteration


Convergence curve
Best fitness
10^5
10^0
10^-5
10^-10
10^-15
0
200
400
600
Iteration